The core challenge: a teacher uploads a scanned PDF of *Romeo and Juliet* or a Rwandan novel, and what the student sees must feel like it was **designed for them** — not like a PDF viewer.

The Problem Visualized

```
WHAT THE TEACHER UPLOADS          WHAT THE STUDENT CURRENTLY SEES

┌──────────────────────┐          ┌──────────────────────┐
│ Romeo and Juliet     │          │ Romeo and Juliet     │
│ Act 1, Scene 1       │          │ Act 1, Scene 1       │
│                      │          │                      │
│ SAMPSON: Gregory,    │    ──►   │ SAMPSON: Gregory,    │
│ o' my word, we'll    │          │ o' my word, we'll    │
│ not carry coals.     │          │ not carry coals.     │
│                      │          │ [Tiny text, no       │
│ GREGORY: No, for     │          │  structure, just a   │
│ then we should be    │          │  wall of words]      │
│ colliers.            │          │                      │
└──────────────────────┘          └──────────────────────┘

        ↓ WHAT IT SHOULD BECOME ↓

┌──────────────────────────────────────────────────────┐
│  Romeo & Juliet  ·  Act 1  ·  Scene 1                │
│  ────────────────────────────────────────────────────│
│  📍 A street in Verona. Two servants enter.          │
│                                                      │
│   ┌──────────┐                                       │
│   │ SAMPSON  │  "Gregory, o' my word, we'll          │
│   │ 🗡 Capulet│  not carry coals."                    │
│   └──────────┘                                       │
│       💬  ┌──────────────────────────┐               │
│           │ What he means: "We won't  │               │
│           │ take insults from anyone." │               │
│           │                          │               │
│           │ "carry coals" = old English │             │
│           │ for accepting humiliation  │               │
│           └──────────────────────────┘               │
│                                                      │
│   ┌──────────┐                                       │
│   │ GREGORY  │  "No, for then we should be           │
│   │ 🗡 Capulet│  colliers."                           │
│   └──────────┘                                       │
│       💬  He's making a joke — "colliers"            │
│           means coal workers. He's playing           │
│           on Sampson's words.                        │
│                                                      │
│  ▶ 🔊 Listen to this scene                           │
│  ▰▰▰▰▰▰▰▱▱▱
```

The PDF hits your backend. Before any student sees it, the ML pipeline must **understand the document's literary structure**.

The Document Intelligence Model:

```
Raw PDF
   │
   ▼
┌──────────────────────────────────────────┐
│         DOCUMENT PARSING LAYER           │
│                                          │
│  1. OCR (if scanned) ─── Tesseract/PaddleOCR│
│  2. Layout Detection ─── Detect columns, │
│     margins, headers, footnotes          │
│  3. Literary Structure Recognition ◄── THIS  │
│     IS THE KEY MODEL                     │
└──────────────────────────────────────────┘
                   │
                   ▼
```

Literary Structure Recognition — a fine-tuned classifier that detects:

Element	Novel	Play	Poem	Short Story
Chapter/Act/Scene breaks	✓	✓	✓ (Stanzas)	✓ (Sections)
Dialogue vs. narration	✓	✓ (all dialogue)	—	✓
Speaker identification	From context	From labels	—	From context
Stage directions	—	✓	—	—
Setting descriptions	✓	✓	—	✓
Internal monologue	✓	✓ (soliloquy)	✓	✓
Verse vs. prose	—	✓ (Shakespeare mixes both)	✓	—

Output: A structured JSON document, not raw text.

```
{
  "type": "play",
  "title": "Romeo and Juliet",
  "author": "William Shakespeare",
  "reading_level": "advanced",
  "structure": [
    {
      "type": "act",
      "number": 1,
      "scenes": [
        {
          "type": "scene",
          "number": 1,
```

```
          "setting": "A public place in Verona",
          "characters_present": ["Sampson", "Gregory"],
          "estimated_read_time_minutes": 8,
          "chunks": [
            {
              "type": "dialogue",
              "speaker": "Sampson",
              "faction": "Capulet",
              "text": "Gregory, o' my word, we'll not carry coals.",
              "difficulty_score": 0.82,
              "archaic_phrases": [
                {
                  "phrase": "carry coals",
                  "modern_meaning": "accept insults or humiliation",
                  "context_note": "This was a common insult in Elizabethan England"
                }
              ],
              "literary_devices": ["wordplay_setup"],
              "emotion": "boastful"
            }
          ]
        }
      ]
    }
  ]
}
```

Stage 2: Pre-Computation — The Book Brain

Before the student reads a single word, the model pre-analyzes the **entire book**:

A. Difficulty Mapping

```
Chapter/Scene Difficulty Curve for "Things Fall Apart"

Difficulty
1.0 |          ┌─┐             ┌──┐
    |         /   \           /    \  ← Ch.13-15
0.8 |   ┌─┐ /     \     ┌─┐/       \    (Okonkwo's exile
    |  /   \/       \   /   \         \   + missionaries arrive)
0.6 | /              \/     \         \
    |/                \       \         \
0.4 |                          \        \
    |                                    \
0.2 |
    └──────────────────────────────────────
     Ch.1  3   5   7   9   11  13  15  17  19
```

For every chunk, pre-compute:

- **Vocabulary difficulty** — word frequency analysis against student's grade level
- **Sentence complexity** — clause depth, passive voice density
- **Conceptual difficulty** — abstract themes, cultural references
- **Predicted struggle points** — specific words/phrases likely to cause problems

B. Character & Relationship Map

Auto-generated and updated as the student progresses:

```
┌──────────┐   father of    ┌──────────┐
│ OKONKWO  │──────────────▶│  NWOYE   │
│ (warrior)│               │  (son)   │
└──────────┘               └──────────┘
     │
     │ kills                    │
     ▼                          │ converts to
┌──────────┐               ┌──────────┐
│ IKEMEFUNA│               │CHRISTIANITY│
│ (foster  │               └──────────┘
│  son)    │
└──────────┘
```

The student can **tap any character name** in the text and see: who they are, their relationships, and a "story so far" for that character. No more "wait, who is this person again?"

C. Cultural Context Bank

For Rwandan students reading Shakespeare, or any cross-cultural text:

> ***"carry coals"*** *— In Shakespeare's time, coal carriers were looked down on. Sampson is saying "we won't be treated like lowly workers." Think of it like someone saying "we won't be disrespected" in Kigali today.*

Pre-generated for every culturally distant reference in the book.

Stage 3: The Adaptive Rendering Engine

This is where the parsed structure becomes a **living reading experience**:

A. Genre-Specific Display Templates

```
┌──────────────────────────────────────┐
│              NOVEL MODE              │
│                                      │
│  Clean paragraph flow                │
│  Character avatars in dialogue       │
│  Chapter progress bar                │
│  "Story so far" recap at chapter start │
│  Vocabulary sidebar (toggleable)     │
└──────────────────────────────────────┘


┌──────────────────────────────────────┐
│              PLAY MODE               │
│                                      │
│  Script-style layout                 │
│  Character cards with faction colors │
│  Stage directions in italic boxes    │
│  Scene visualizer (who is on stage?) │
│  "What just happened" after each scene │
└──────────────────────────────────────┘


┌──────────────────────────────────────┐
│              POEM MODE               │
│                                      │
│  Centered verse with breathing room  │
│  Line-by-line audio playback         │
```

```
│  Rhyme scheme highlighting                │
│  Meaning layer (tap to reveal)            │
│  Emotional tone color gradient            │
└──────────────────────────────────────────┘
```

B. The Dyslexia Rendering Layer

Every text chunk passes through:

```
Raw text chunk
      │
      ▼
┌──────────────────────────────────────┐
│  DYSLEXIA RENDERING PIPELINE         │
│                                      │
│  1. Font → OpenDyslexic / Lexie      │
│  2. Line spacing → 1.8x minimum      │
│  3. Letter spacing → +2px            │
│  4. Line length → max 60 chars       │
│  5. Paragraph chunking → 3-4 lines   │
│  6. Syllable markers on hard words    │
│  7. Alternating line backgrounds      │
│     (subtle gray/white bands)         │
│  8. Reading ruler follows scroll      │
└──────────────────────────────────────┘
```

C. The ADHD Engagement Layer

```
Full chapter (20 pages)
      │
      ▼
┌──────────────────────────────────────┐
│  ADHD CHUNKING ENGINE                │
│                                      │
│  Break into 2-3 minute segments      │
│  Each segment ends with:             │
│    → Micro-quiz (1 question)         │
│    → Progress reward animation       │
│    → "What happens next" teaser      │
│                                      │
│  Between segments:                   │
│    → 10-second breathing break       │
│    → Choice: continue or bookmark    │
│                                      │
│  Gamification:                       │
│    → Reading streak counter           │
│    → Chapter completion badges        │
│    → "You understood 8/10 scenes!"   │
└──────────────────────────────────────┘
```

Stage 4: The Live Highlight-to-Understand Feature

The flagship interaction — student highlights anything, gets instant help:

```
Student highlights: "we'll not carry coals"
                            |
                            ▼
              ┌──────────────────────────┐
              |   CONTEXT ANALYZER       |
              |                          |
              | What was highlighted?    |
              | → Idiomatic expression   |
              |                          |
              | Where in the book?       |
              | → Act 1, Scene 1, line 1|
              | → Sampson speaking       |
              | → Pre-fight tension      |
              |                          |
              | Student's level?         |
              | → Reading level: Grade 4 |
              | → Vocabulary: developing |
              | → Prefers: examples      |
              └──────────┬───────────────┘
                         |
                         ▼
              ┌──────────────────────────┐
              |   SIMPLIFICATION LLM     |
              |   (book-aware, student-  |
              |    aware)                |
              └──────────┬───────────────┘
                         |
                         ▼
     ┌──────────────────────────────────┐
     |                                  |
     |  "We won't take insults"         |
     |                                  |
     |  ────────────────────────────    |
     |  "Carry coals" is an old English |
     |  expression. Coal carriers were   |
     |  seen as low-status workers.     |
     |                                  |
     |  Sampson is saying: "Nobody messes |
     |  with us." He's trying to sound  |
     |  tough in front of Gregory.      |
     |                                  |
     |  🔊 Hear it   📖 See it used again |
     |  ✏️ Add to my vocabulary list     |
     └──────────────────────────────────┘
```

Key design rules:

1. **First line: the simple meaning** — always. No student should have to read a paragraph to get the point.
2. **Second block: the why** — context, author intent, cultural background. Optional to read.
3. **Action buttons** — hear it, save it, see other examples. Student controls depth.
4. **Never break immersion** — the explanation appears as an overlay, not a new page. One tap to dismiss and keep reading.

Stage 5: The Feedback Loop

Every interaction feeds back into the model:

```
Student highlights "carry coals"
        │
        ├──▶ Log: vocabulary_gap detected
        │    (archaic_idiom, Elizabethan English)
        │
        ├──▶ Update student embedding:
        │    idiom_comprehension -= 0.1
        │
        ├──▶ RL Agent receives signal:
        │    "Pre-teach idioms before next scene"
        │
        └──▶ Teacher dashboard:
             "3 students struggled with Elizabethan
              idioms in Act 1. Consider a 5-min
              vocabulary warm-up before Act 2."
```

The Complete Pipeline

```
  TEACHER UPLOADS PDF
        │
        ▼
  ┌──────────────┐
  │  OCR + Parse  │ ← Extracts raw text from any PDF
  └──────────────┘
        │
        ▼
  ┌──────────────┐
  │  Literary     │ ← Identifies: novel/play/poem,
  │  Structure    │   chapters, dialogue, characters,
  │  Recognition  │   stage directions, verse/prose
  └──────────────┘
        │
        ▼
  ┌──────────────┐
  │  Book Brain   │ ← Pre-computes: difficulty map,
  │  Pre-Analysis │   character graph, cultural notes,
  │               │   vocabulary list, predicted struggles
  └──────────────┘
        │
        ▼
  ┌──────────────┐
  │  Structured   │ ← Clean JSON: every sentence tagged
  │  Content DB   │   with metadata, difficulty, context
  └──────────────┘
        │
        ▼
  ┌──────────────────────────────────────┐
  │        STUDENT READING SESSION        │
  │                                       │
  │  Genre template (novel/play/poem)     │
  │  + Dyslexia rendering layer           │
  │  + ADHD engagement layer              │
  │  + Live highlight-to-understand       │
  │  + RL agent adjusting in real-time    │
  │  + Comprehension tracking             │
```

```
└──────────────────────────────────┘
        │
        ▼
┌──────────────┐
│  Teacher gets │ ← "Here's what your class
│  insights     │    struggled with today"
└──────────────┘
```

What to Build First

Priority	Component	Why first
1	PDF parser + literary structure recognition	Nothing works without clean structured content
2	Genre-specific display templates	Students must see clean, accessible text immediately
3	Highlight-to-understand (simplification LLM)	Highest impact single feature — students self-serve
4	Dyslexia rendering layer	Font, spacing, syllable markers — measurable impact
5	ADHD chunking engine	Breaks content into digestible pieces
6	Book Brain pre-analysis	Difficulty maps, character graphs, cultural context
7	RL policy for real-time adaptation	Ties everything together intelligently